HAT-P-12b

R-0852

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-12_180405` · created 2026-07-28T20:53:18+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458213.8216 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.01
Transit depth (Rp/Rs)^2	2.75 +/- 0.34 [%]
Orbital Inclination (inc)	87.79 +/- 0.77
Airmass coefficient 1 (a1)	1.189 +/- 0.018
Airmass coefficient 2 (a2)	-0.159 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	4.73
Optimal Annulus	11.83
Transit Duration (day)	0.0925 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.01 vs 0.1406 (deviation 2.39σ)
Duration fit vs published	0.0925 d vs 0.0975 d (deviation 0.78σ)
Mid-time O–C	5.7 min
Depth SNR	15.6
Residual scatter	1.47 %

Light curve

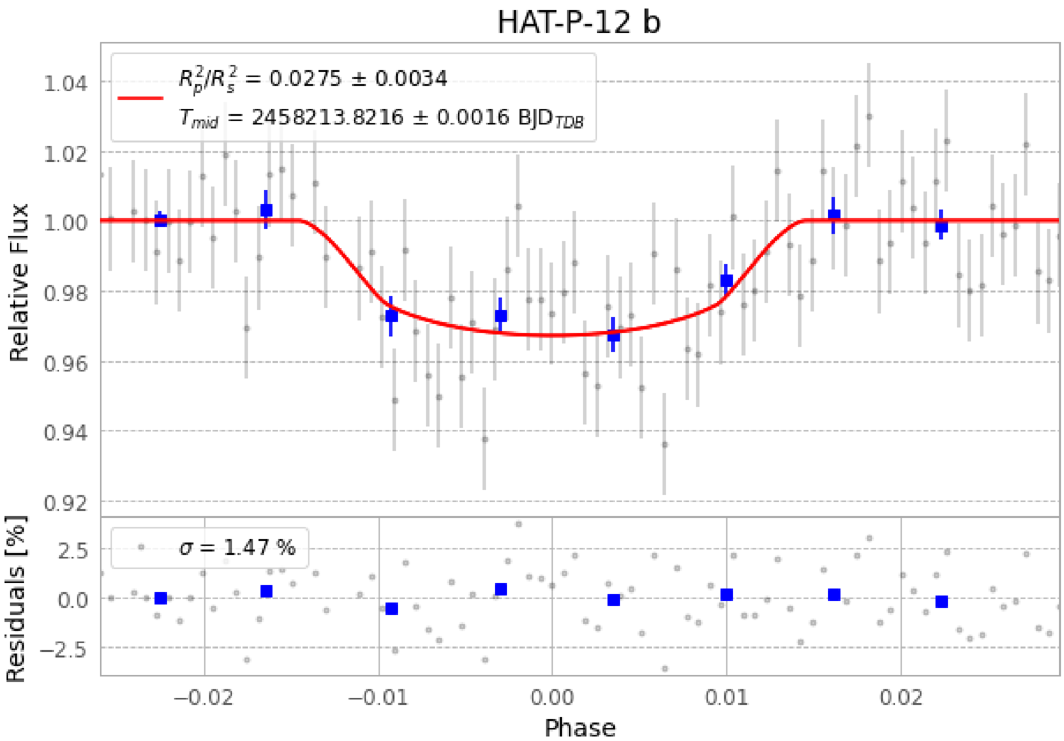

Figure 1 — FinalLightCurve_HAT-P-12 b_2018-04-04.png (SHA-256 8c51cf79b8428a05…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 232], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5171bf108a4f93b2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

84 FITS frames (55 MB), `HATP-12180405053612.FITS` → `HATP-12180405095112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-180405.log (a8eb0f4761ac…), FinalLightCurve_HAT-P-12 b_2018-04-04.png (8c51cf79b842…), FinalLightCurve_HAT-P-12 b_2018-04-04.csv (d8e2a162c186…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)